Capstone Project — Technical Profile (Updated)

Section 1: Basic Information

Item	Description
Student Name	**Ant Bone Kyaw, Hnin Ei Ei Aung, Thaw Zin Moe Myint**
Student ID	66130500802, 66130500808, 66130500824
Program / Major	Bachelor of Science in Computer Science
Project Title	**SmartFin: Automatic Expense Tracking with Thai E-Slip Recognition and an AI Financial Coach**
Project Type	☒ Application / System Development ☐ Research / Innovation. ☐ Mixed

Section 2: Technical Profile — Knowledge and Application Domains (Fill only the relevant domains or adjust as suitable for your project)

Technical Domain	Key Focus / Example Topics	Tools / Technologies	Current Proficiency *	Planned Action (Use / Learn new / Enhance / Innovate)	Expected Role / Application in Project
Programming & Software Development	Cross-platform mobile app, REST API, device features (camera, photo library, notifications)	Flutter (Dart), Go (Fiber), photo_manager, local notifications	Advanced	Enhance	Built the Flutter app for Android and iOS and the Go API for transactions, monthly summaries, bills and the AI Coach
Data, AI & Analytics	Reading Thai e-slips and physical receipts; agentic RAG with function calling	Google Gemini API (Gemini 3.5 Flash-Lite), Google Gen AI Go SDK	Intermediate	Learn New / Innovate	Gemini Vision reads merchant, Baht amount and date, converting Buddhist-era years; the AI Coach picks among 7 tools, so its figures come from the user's own records
Database & Storage	Document schema, aggregation pipelines, indexes	MongoDB Atlas, MongoDB Go driver, GridFS	Intermediate	Enhance	Stores transactions, monthly summaries, bills and chat history per user; aggregations drive the budget and category views
Infrastructure & DevOps	Cloud hosting, version control, managed services	Google Cloud Run, GitHub, Firebase Auth, Postman	Intermediate	Learn New	API hosted on Google Cloud Run (Singapore); GitHub repository; Firebase sign-in; Postman API tests
UX/UI & Interaction Design	Design system, readable contrast, confirm-before-save flow	Figma, Flutter theming	Intermediate	Enhance	One UI kit across every screen; the user confirms every AI-read slip, with a warning before saving a duplicate

Technical Domain	Key Focus / Example Topics	Tools / Technologies	Current Proficiency *	Planned Action (Use / Learn new / Enhance / Innovate)	Expected Role / Application in Project
Applied / Research Domain	Personal finance (FinTech): ground-truth evaluation of LLM answers	Go evaluation tools, rule-based scoring against MongoDB	Intermediate	Innovate	Tests the AI Coach against one-shot RAG and no-retrieval baselines on 28 tasks, and slip reading on 30 real e-slips and receipts

*Proficiency: Beginner / Intermediate / Advanced

Section 3: Summary Reflection

1. **Key knowledge or skills to enhance:**
   Go backend with Fiber and MongoDB aggregation; prompting Gemini to extract data from Thai-script images; a function-calling agent with a 45-second time budget; Flutter plugins for the photo library and notifications; Thai time (UTC+7) and Buddhist-era dates.
2. **Enhanced or innovative elements planned to develop:**
   Slip-to-ledger without typing: photograph an e-slip or physical receipt, or let Auto-Detect find new e-slips in the gallery; the user confirms before saving.
   Agentic RAG AI Coach: answers in Thai or English from the user's own records; can set a budget or log an expense.
   Bill tracker: on-device reminders for recurring payments, guarded against paying twice.
3. **Anticipated technical risks or challenges:**
   Thai script, Buddhist-era years and rotated photos: explicit prompt plus a mandatory verify screen.
   Made-up figures from the coach: answers use only tool results; every Baht amount is checked in testing.
   API cost and rate limits: free-tier Flash-Lite model and a bounded agent loop.
   Remaining risk: Firebase sign-in tokens are not yet verified by the API; needed before public release.
4. **How to validate technical outcomes (proof, testing, etc.):**
   Slip reading: 30 real e-slips and physical receipts, scanned twice: amount correct in 100% of scans, all fields in 90%.
   AI Coach: 28 scripted tasks × 3 runs, checked against figures recomputed from MongoDB: 94% success vs 87% (one-shot RAG) and 18% (no data).
   Software testing: Go unit and Flutter widget tests, test design specification and test cases, alpha test (14 Sep 2026) and beta test; approved for D-Day on 20 Oct 2026.

Section 4: Advisor Acknowledgment

Aspect	Advisor Comment
Appropriateness of Enhancement / Innovation	
Practical/Research consideration	

Reviewed by: _____ Date: ____ / ____ / ____

(Advisor)